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Theoretical and experimental NMR studies on muscimol from fly agaric mushroom (*Amanita muscaria*)

Teobald Kupka* and Piotr P. Wieczorek*

Faculty of Chemistry, Opole University, Oleska 48, 45-052 Opole, Poland

Abstract

In this article we report results of combined theoretical and experimental NMR studies on muscimol, the bioactive alkaloid from fly agaric mushroom (*Amanita muscaria*). The assignment of ^{1}H and ^{13}C NMR spectra of muscimol in DMSO-d_6 was supported by additional two-dimensional heteronuclear correlated spectra (2D NMR) and gauge independent atomic orbital (GIAO) NMR calculations using density functional theory (DFT). The effect of solvent in theoretical calculations was included via polarized continuum model (PCM) and the hybrid three-parameter B3LYP density functional in combination with 6-311++G(3df,2pd) basis set enabled calculation of reliable structures of non-ionized (neutral) molecule and its NH and zwitterionic forms in the gas phase, chloroform, DMSO and water. GIAO NMR calculations, using equilibrium and rovibrationally averaged geometry, at B3LYP/6-31G* and B3LYP/aug-cc-pVTZ-J levels of theory provided muscimol nuclear magnetic shieldings. The theoretical proton and carbon chemical shifts were critically compared with experimental NMR spectra measured in DMSO. Our results provide useful information on its structure in solution. We believe that such data could improve the understanding of basic features of muscimol at atomistic level and provide another tool in studies related to GABA analogs.

Keywords: muscimol; NMR spectroscopy; DFT; GIAO NMR

Correspondence to: teobaldk@gmail.com (TK) or Piotr.Wieczorek@uni.opole.pl (PW)
Faculty of Chemistry, Opole University, Oleska 48, 45-052 Opole, Poland
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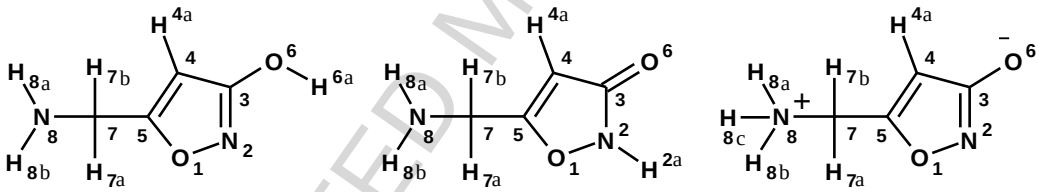
The close coexistence of humans and beautifully colored red fly agaric mushroom with white spots (*Amanita muscaria*) in the green forest has long lasting history. One can see it in art, medicine, children literature, as well as in the use of the mushroom as psychotropic "inspiration" in religious and magical practices[1,2]. The genus *Amanita* produces several alkaloids including muscimol, ibotenic acid, muscazone and muscarine which react with neurotransmitter receptors in the central nervous system. The first three alkaloids belong to the chemical group of isoxazoles[3]. Apart from its remarkable appearance, the mushroom significance is related to two main isoxazole alkaloids – ibotenic acid and muscimol[3-7]. Muscazone and muscarine exhibit lower hallucinogenic activity than ibotenic acid and muscimol[3]. Molecular structures of main halucinogenic substances, both ibotenic acid and muscimol, are similar to human neurotransmitters: glutamic acid and GABA[7,8] (γ-aminobutyric acid). Muscimol[9] (5-(aminomethyl)isoxazol-3-ol) is a degradation product of ibotenic acid. Three potential forms of muscimol are shown in **Scheme 1**:



(A) non-ionized (neutral) (B) NH form (C) Zwitterion

Scheme 1 Three forms of muscimol with atom numbering.

Muscimol contains a disubstituted isoxazole ring and the zwitterionic form of its molecule was postulated[6,9]. About 25 years ago appeared an excellent review "The chemistry of heterocyclic compounds; Isoxazoles[10]". The reader is referenced to this book for basic information concerning synthesis[11-17], structure[9], properties and some spectroscopic data[10] of this important group of heterocyclic compounds. On the other hand, some information in this review, including theoretical calculations and instrumental analysis are outdated. Recently, synthesis of several isoxazoles as acetylcholinesterase inhibitors were reported[18,19]. The isoxazole ring is very popular[19,20] and is also present in some semi-synthetic penicillins, for example in cloxacillin[10,21].

Currently, the most interesting biological feature[7,22] of muscimol is its action on (human) central nervous system[2,23]. Thus, muscimol is a potent GABA[6,24] agonist. As result, it has

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been widely applied in biological, physiological, behavioral and medical studies[23,25]. Many GABA agonists and antagonists have been studied, too[19,25-30]. The idea behind these studies is the similarity of GABA and muscimol flexible molecular structure and characteristic dipolar electron density distribution[6,25,30,31]. Both molecules contain two, well separated ends, capable of interaction with specific brain receptors via terminal groups bearing opposite charges[6]. Thus, the molecular size, flexibility and charge distribution seem to be important features of GABA analogs[5,6,20,25,30-32]. Besides, GABA has been described as one of the most important brain metabolites non-invasively observed in vivo by magnetic resonance spectroscopy[33]. Its proton NMR spectra[33,34] are currently widely used in localized NMR spectroscopy of brain, including studies of visual stimulation in fNMR protocols[35]. In addition, some modern applications of muscimol include its immobilization on modified silicon surfaces[36].

Recently, some of us proposed a rapid and simple NMR method of muscimol determination in the presence of ibutenic acid in solution (including body fluids[37]). Understanding the important role of muscimol and other GABA analogs in brain activity we were surprised noticing only very few and old basic chemical studies on muscimol in the literature[10,19]. Obviously, there have been several reports on extraction of muscimol from the mushroom and its synthetic routes in the laboratory, as well as its qualitative and quantitative determination[2,13,19,38-42]. There were also some early ^{1}H and ^{13}C NMR studies of muscimol in solution[10,11]. Unfortunately, it is not easy to find the original sources and verify these data with modern approach. Several reports also described application of sensitive chromatographic, and other analytical methods, for determination of muscimol[2,3,11,39,40]. Obviously, older molecular modeling studies on GABA, ibutenic acid and muscimol structures were performed at semi-empirical and Hartree-Fock levels of theory[31,43,44]. Some of these theoretical works, combined with fairly small basis sets, supported crystallographic works[9,31] and analyzed charge distribution within muscimol molecule.

According to the literature[9,31,45], muscimol is a polar molecule, easily soluble in water, ethyl alcohol and DMSO. However, what is the molecular structure of muscimol in solution and in the crystal state? X-ray studies[9,31] indicate the presence of two molecules of muscimol in its zwitterionic form and water in the elemental cell, held together by strong intermolecular hydrogen bonds (see the three forms of muscimol molecule and the atom numbering in Schemes 1A - C). On the other hand, the problems in proper localization of very light hydrogen atoms by X-ray method are well known. For example, the electron diffraction (ED) or microwave (MV) determined length of a typical C-H bond in the gas phase[46] is about 1.09 Å but from single crystal studies[47] this interatomic distance is reported as 0.9 to 0.96 Å.

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The problem with presence of zwitterion form in the gas phase and solution was tackled theoretically and experimentally in case of single GABA molecule and its complexes with a number of water molecules[48-50]. From these studies emerges that in the presence of water GABA could exist as zwitterion[51-53]. However, this form is only slightly more stable than the neutral form and the results are strongly method and basis set dependent. For example, at B3LYP/6-311++G** level of theory the canonical form of GABA + 5 water molecules was found slightly more stable than the zwitterion one[48]. ^{1}H NMR parameters of non-ionized and zwitterionic GABA were also calculated theoretically[51].

Besides, the majority of reports present muscimol formula as non-ionized neutral molecule (Scheme 1A, see also Fig. 1 in ref. [32]). The further complication is due to a possibility of migration of hydrogen atom from a hydroxyl group to nitrogen atom in isoxazole ring (Scheme 1B). In fact, the two latter forms resemble the canonic keto (CK) and enol, or amino-hydroxy (CE) forms of cytosine which is dominating in solution[54-57]. Recently, the structure and charge distribution of GABA and muscimol in the gas phase and solution was studied in details theoretically by Serdaroglu[58].

The primary goal of this paper is to elucidate the structure of muscimol in solution using a combined experimental and theoretical approach. To this end, the stability of the three analyzed muscimol forms in the gas phase and chloroform, DMSO and water solutions were evaluated on the basis of fully unconstrained geometry optimization and comparison of the calculated total energy. We were particularly interested in comparing our NMR measurements with the results of theoretical calculations. ^{1}H and ^{13}C NMR spectroscopy in DMSO-d6 solution of muscimol was selected as source of experimental information of muscimol molecular structure in solution. The current approach is partly similar to recent theoretical and experimental ^{1}H NMR studies on GABA[51], as well as cytosine and 1-methylcytosine in solution[54]. However, in our study the analysis of the observed NMR data was supported by high-level DFT calculations combined with fairly complete basis sets and using both equilibrium and rovibrationally averaged geometry[59-62]. Both isotropic nuclear magnetic shieldings and chemical shifts relative to TMS in the gas phase and solution using polarized continuum model (PCM[63]) were calculated within gauge independent atomic orbital (GIAO[64-66]) NMR approach using 6-31G*[67] and aug-cc-pVTZ-J[68-70] basis set. The latter basis set was originally tailored for prediction of indirect spin-spin coupling constants. However, later it was reported to produce also accurate nuclear shieldings (close to the complete basis set limit) for several small model molecules[71].

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2. Computational and experimental details

2.1. Theoretical approach

Three models of muscimol molecule were created with GaussView program[72] (see the corresponding atom numbering in Scheme 1). Gaussian 09[62] software was used in fully unconstrained geometry optimization of these molecules, followed by vibrational analysis and NMR calculations. The harmonic model of vibration calculations was selected to make sure the optimization produced minimum energy structures. Obviously, in accurate calculations of molecules with multiple bonds and lone electron pairs the use of large basis sets, including both polarization and diffuse functions was essential. A versatile and generally well performing B3LYP hybrid density functional[73-75], combined with 6-311++G** and 6-311++G(3df,2pd) basis sets, was used to fully optimize the three models of free muscimol at equilibrium geometry (R_e) in the gas phase, and in $CHCl_3$, DMSO and water solutions. The solvent effect on selected properties of dissolved muscimol was modeled using the self-consistent reaction field (SCRF) calculations within the polarized continuum model (PCM[63]). The Mulliken and electrostatic potential fitted charges on all atoms, and with hydrogen summed on "heavy" atoms, were calculated at B3LYP/6-311++G(3df,2pd) level of theory in the gas phase and in water.

Finally, the B3LYP/6-311++G(3df,2pd) calculated equilibrium (R_e) muscimol structures in the gas phase and in solution were used for the subsequent prediction of nuclear shieldings using the gauge independent atomic orbital (GIAO) approach[65,66]. The GIAO NMR parameters are very sensitive to the completeness and quality of the used basis set[67,76-78]. Thus, for calculation of nuclear magnetic shieldings and indirect spin-spin coupling constants we selected the B3LYP/aug-cc-pVTZ-J level of theory. The Sauer and coworkers' aug-cc-pVTZ-J basis set[68-70] (downloaded from Environmental Molecular Sciences Laboratory exchange basis set library (EMSL[79,80]). was selected for calculation of nuclear shieldings. In our earlier studies we noticed its good performance in predicting proton and carbon NMR shieldings[71,81].

Some rovibrational structure (R_v) optimizations[62], using the second order vibrational perturbational theory (VPT2[82-85]), combined with NMR shielding calculations, were also performed at lower level of theory (B3LYP/6-31G* and B3LYP/6-311++G**) in the gas phase and DMSO. Obviously, these calculations were very demanding computationally for a molecule of medium size (attempts with larger basis sets failed).

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Theoretical chemical shifts (δ in ppm) were referenced to tetramethylsilane (TMS) nuclear shieldings (σ) calculated at the same level of theory, and the corresponding carbon and proton parameters were calculated as follows:

$$\delta(C_i \text{ or } H_i) = \sigma(TMS) - \sigma(C_i \text{ or } H_i) \quad \textbf{(1)}$$

These formulas were also used to derive experimental nuclear shieldings of muscimol. In this case the experimental shieldings of liquid TMS from the literature[86,87] were used:

$$\sigma(C_i) = 186.37 - \delta(C_i) \quad \textbf{(2)}$$

$$\sigma(H_i) = 32.815 - \delta(H_i) \quad \textbf{(3)}$$

The rovibrational corrections to nuclear shieldings[59,60,88], e. g. zero-point vibrational corrections (ZPVC) and temperature corrections (TC) were calculated at B3LYP/6-31* level of theory using VPT2 method, both in the gas phase and in DMSO. This allowed theoretically more justified comparison of the predicted results with experiment (σ (Exp.)). Thus, the empirical values[88-90] σ (Emp.) were derived from experimental shieldings (calculated from eq. 3 and 4) augmented with rovibrational corrections:

$$\sigma\,(Emp.) = \sigma\,(Exp.) - (ZPVC + TC) \quad \textbf{(4)}$$

Theoretical carbon-proton indirect spin-spin coupling constants (SSCC) were also calculated in Gaussian program. Due to difficulties in calculating the corresponding ZPV corrections only the equilibrium SSCC values will be discussed.

The accuracy of theoretical predictions is often expressed by several statistical indexes, including deviation from experiment, mean value of deviation and the root-mean-square (RMS) deviation from experimental values. Several experimental muscimol bond lengths for "heavy or non-hydrogen" atoms were available from the literature. However, only 4 carbon and 2 proton chemical shifts from our measurements were available for statistics. So, from statistical point of view, the NMR data should be discussed in terms of averaged deviations of calculated values from experiment. However, for consistency with structural parameters, we decided to use RMS as a total rough measure of theoretical prediction quality in the current study. In this work we applied the following RMS formula:

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$$RMS = \sqrt{\frac{[(x_i - x_{i_{\exp}})^2 + ...(x_n - x_{n_{\exp}})^2]}{n}} \quad \textbf{(5)}$$

2.2. NMR experiment

Muscimol was supplied from Abcam Biochemicals, Cambridge. Bruker Ultrashield Avance 400 NMR spectrometer, operating at 400.13 and 100.613 MHz for ^{1}H and ^{13}C NMR, was used for room-temperature measurements (293K). Standard TOPSPIN ver. 1.3 software was used for acquiring and processing NMR signals.

About 2.5 mg of muscimol was dissolved in 0.5 ml of DMSO-d_6 with TMS added as internal reference and placed in 5 mm o. d. NMR tube. The accuracy of proton and carbon chemical shift measurements was ±0.001 and ±0.01 ppm (or better). Internal solvent lock was used for magnetic field stabilization. 256 signals were accumulated using fairly long acquisition time (AT=4.0 s) and short delay (D1=0.5 s) for routine 1D proton spectra. For better signal resolution no line broadening (LB=0) was used during signal processing. Typical one-dimensional proton decoupled spectra (^{13}C {-^{1}H} NMR) were recorded using 17500 scans, each of 2 s duration and separated with 1s delay. Specific line broadening, both for sensitivity enhancement (LB=5 Hz) and better resolution (LB=0) were selected. Due to low concentration significantly higher number of acquisitions was set for gated proton coupled carbon spectra (^{13}C {+^{1}H} NMR, NS=65500, AT=2.0 s, D1=1.0 s). Besides, in order to read small J couplings, no line broadening prior Fourier transformation was selected.

For unequivocal signal assignment additional two-dimensional (2D) spectra were measured[91]. In the first case a correlation between directly connected proton-carbon pairs was evident from a Heteronuclear Single Quantum Correlation (HSQC) spectrum. The spectrum was recorded using the following acquisition parameters: NS=128, AT=0.075 and D1=2.0 s. A coupling between proton and carbon atoms separated by 2-4 bonds was clearly visible (one-bond cross-peaks were partly suppressed) from a Heteronuclear Multiple Bond Correlation (HMBC or long-range correlation) spectrum of muscimol recorded using NT=256, AT=0.151 and D1=1.5 s, respectively.

3. Results and discussion

3. 1. Muscimol structure in the gas phase and solution.

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We will start our discussion with a presentation of selected B3LYP/6-311++G(3df,2pd) calculated structural parameters of the three studied molecular forms of muscimol in the gas phase and solution (Table 1). To show the numerical impact of solvent polarity we also performed structural optimization in three solvents including chloroform. However, despite numerous attempts we failed to dissolve muscimol in chloroform in concentration sufficient to measure its ^{1}H NMR spectrum in an overnight experiment (e. g., the assumed concentration was below 0.0001 mg/ml).

According to the literature[9], no gas phase structure of muscimol is available. However, there were several single crystal X-ray studies[9] on muscimol molecular structure at room (294 K) and low temperature (122 K). Its crystal lattice contained two muscimol molecules and one water. Obviously, the X-ray technique cannot precisely locate light atoms[47] (here hydrogen) and we will only discuss interatomic distances and angles between heavier atoms (C, N and O). According to these X-ray studies, the two muscimol molecules (A and B) in the elementary cell are in zwitterionic form and are stabilized by strong intermolecular hydrogen bonds between positively charged NH_3^+ end group in molecule A and a negative substituent at the opposite side ($=O^-$) of molecule B.

We are aware that the inclusion of polar environment effect via a simplified solvent model (e. g., without a possibility of producing specific hydrogen bonding network between solute and solvent) is a fairly rough approximation. However, in many theoretical studies this model is quite sufficient. Thus, by analogy to the crystal state, our theoretical structures in polar solvent should better resemble the zwitterionic form than the results of gas phase optimization.

At this point we will briefly discuss the possibility of intramolecular bond formation between O1 and the closest H atom from an amino group (see Scheme 1). The experimental O1 … N8 separation[9] of 3.057 Å at 122K is accurately reproduced by our calculations (see Tab. 1). The calculated (N8)H8b …O1 separations for NH, neutral and zwitterion forms of muscimol in the gas phase (2.831, 2.846 and 1.951 Å) and in water (2.842, 2.872 and 2.713 Å) suggest a presence of very weak H-bond contacts. Assuming a separation below 2 Å for strong H-bonds, this suggests a presence of fairly small intramolecular H-interactions in the discussed structures in solution. Obviously, direct hydration should produce significantly stronger solute-solvent interactions.

Looking at the results gathered in Table 1 we immediately notice big differences in calculated C3-O6 bond lengths (marked with bold font) for NH, neutral and zwitterion forms of

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muscimol. The C3-O6 bond lengths for structures optimized in the gas phase are about 1.21, 1.34 and 1.24 Å and in water are longer by about 0.02 Å (1.23, 1.34 and 1.26 Å).

Besides, the corresponding dihedral angle (O1C5C7N8), representing the arrangement of $CH_2NH_{2(3)}$ chain with respect to the isoxazole ring, is also sensitive to the three molecular forms of muscimol and the nature of solvent (about 70, 60 and 90 degrees in the gas phase and 70, 70 and 67 degrees in water).

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Table 1 Selected B3LYP/6-311++G(3df,2pd) calculated interatomic distances (in Å) and dihedral angle (in deg) in NH, neutral and zwitterion forms of muscimol in the gas phase and in solution[a]

Bond and angle	Gas	$CHCl_3$	DMSO	Water
	NH form			
O1-N2	1.414	1.408	1.406	1.406
N2-C3	1.417	1.405	1.401	1.401
C3-C4	1.455	1.450	1.447	1.447
C4-C5	1.345	1.347	1.348	1.348
C5-O1	1.358	1.357	1.356	1.356
C3-O6	**1.211**	**1.220**	**1.224**	**1.225**
C5-C7	1.497	1.497	1.497	1.497
C7-N8	1.462	1.464	1.465	1.465
O6-H	-	-	-	-
N8-Haver	1.013	1.013	1.013	1.013
C4-H	1.075	1.075	1.075	1.075
(N8)H8b-O1	2.831	2.841	2.842	2.842
N8-O1	3.059	3.067	3.070	3.070
N2-H	1.012	1.012	1.012	1.012
Dihedral angle				
O1C5C7N8	69.545	70.257	70.553	70.568
	Neutral			
O1-N2	1.406	1.405	1.406	1.406
N2-C3	1.308	1.308	1.308	1.308
C3-C4	1.422	1.419	1.419	1.419
C4-C5	1.360	1.356	1.356	1.356
C5-O1	1.350	1.348	1.349	1.349
C3-O6	**1.342**	**1.340**	**1.339**	**1.339**
C5-C7	1.486	1.497	1.496	1.496
C7-N8	1.470	1.466	1.467	1.467
O6-H	0.966	0.967	0.967	0.967
N8-Haver	1.013	1.013	1.013	1.013
C4-H	1.074	1.074	1.074	1.074
(N8)H8b-O1	2.846	2.864	2.872	2.872
N8-O1	2.913	3.086	3.097	3.098
Dihedral angle				
O1C5C7N8	59.604	68.760	70.052	70.136
	Zwitterion			
O1-N2	1.399	1.444	1.437	1.437
N2-C3	1.298	1.353	1.349	1.349
C3-C4	1.502	1.471	1.465	1.464
C4-C5	1.329	1.341	1.343	1.343
C5-O1	1.451	1.345	1.346	1.346
C3-O6	**1.238**	**1.255**	**1.262**	**1.263**
C5-C7	1.540	1.482	1.484	1.484
C7-N8	1.470	1.522	1.515	1.515
O6-H	-	-	-	-
N8-Haver	1.036	1.022	1.021	1.021
C4-H	1.076	1.076	1.076	1.076
(N8)H8b-O1	1.951	2.500	2.705	2.713
N8-O1	3.562	2.866	2.983	2.987
Dihedral angle				
O1C5C7N8	90.138	58.733	66.548	66.790

a) results for C3-O6 (the most sensitive parameter) are marked in bold font.

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Table 2 Selected experimental bond distances and dihedral angle (in Å and deg) of muscimol, the corresponding B3LYP/6-311++G(3df,2pd) calculated deviations from experiment and RMS values for NH, neutral, and zwitterion forms in the gas phase and solution[a]

Parameter	Exp[b]. X-ray	Lit.[c] HF/6-31+G*	Lit[c].	Deviation from Exp. Gas	$CHCl_3$	DMSO	Water
			NH form				
O1-N2	1.425			-0.011	-0.017	-0.019	-0.019
N2-C3	1.314			0.103	0.091	0.087	0.087
C3-C4	1.437			0.018	0.013	0.010	0.010
C4-C5	1.345			0.000	0.002	0.003	0.003
C5-O1	1.356			0.002	0.001	0.000	0.000
C3-O6	**1.303**			**-0.092**	**-0.083**	**-0.079**	**-0.078**
C5-C7	1.489			0.008	0.008	0.008	0.008
C7-N8	1.492			-0.030	-0.028	-0.027	-0.027
N8-O1	3.057			0.002	0.010	0.013	0.013
O1C5C7N8	71.1			-1.55	-0.843	-0.547	-0.532
RMS				**0.048**	**0.043**	**0.041**	**0.041**
			Neutral				
O1-N2	1.425			-0.019	-0.020	-0.019	-0.019
N2-C3	1.314			-0.006	-0.006	-0.006	-0.006
C3-C4	1.437			-0.015	-0.018	-0.018	-0.018
C4-C5	1.345			0.015	0.011	0.011	0.011
C5-O1	1.356			-0.006	-0.008	-0.007	-0.007
C3-O6	**1.303**			**0.039**	**0.037**	**0.036**	**0.036**
C5-C7	1.489			-0.003	0.008	0.007	0.007
C7-N8	1.492			-0.022	-0.026	-0.025	-0.025
N8-O1	3.057			-0.144	0.029	0.040	0.041
O1C5C7N8	71.1			-11.496	-2.340	-1.048	-0.964
RMS				**0.051**	**0.021**	**0.022**	**0.023**
			Zwitterion				
O1-N2	1.425	1.455	0.03	-0.026	0.019	0.012	0.012
N2-C3	1.314	1.336	0.022	-0.016	0.039	0.035	0.035
C3-C4	1.437	1.493	0.056	0.065	0.034	0.028	0.027
C4-C5	1.345	1.321	-0.024	-0.016	-0.004	-0.002	-0.002
C5-O1	1.356	1.324	-0.032	0.095	-0.011	-0.010	-0.010
C3-O6	**1.303**	**1.223**	**-0.08**	**-0.065**	**-0.048**	**-0.041**	**-0.040**
C5-C7	1.489	1.490	0.001	0.051	-0.007	-0.005	-0.005
C7-N8	1.492	1.516	0.024	-0.022	0.030	0.023	0.023
N8-O1	3.057	2.573	-0.484	0.505	-0.191	-0.074	-0.070
O1C5C7N8	71.1	43.2	-27.9	19.038	-12.368	-4.552	-4.310
RMS			**0.166**	**0.176**	**0.069**	**0.033**	**0.032**

a) In bold are marked results for the characteristic C3-O6 bond and RMS values; b) from ref.[31]; c) For comparison are shown old theoretical data from ref.[31]

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The deviations of theoretical geometric parameters of the three potential forms of muscimol from earlier reported X-ray data[31] are compared in Table 2. In particular, the smallest values of RMS deviations should allow us to select the molecular form of muscimol corresponding to X-ray data. In addition, in Table 2 are included early theoretical data from the literature[31], calculated at very low level of theory (HF/6-31+G*). For completeness, in the supplementary material we will place the optimized muscimol geometries. As expected, it is apparent from Table 2 that the gas phase optimized structures produce the largest RMS values (0.05, 0.05 and 0.17 Å for NH, neutral and zwitterion forms). However, in a polar solvent a significant drop of RMS is visible (0.04, 0.02 and 0.03 Å, respectively). In addition, the most pronounced disagreement between theory and experiment (the largest deviation of about 0.08 Å between theoretical and X-ray determined C3-O6 bond length) is observed for the NH form of muscimol in solution. In addition, the HF/6-31+G* results from the literature show a very poor agreement with crystal structure of muscimol (RMS of 0.17 Å). In conclusion, the optimized NH form of muscimol is the least similar one to the experimental data in the crystal state.

In Table 3 are gathered the values of calculated total energies and dipole moments for the three studied forms of muscimol in the gas phase and in chloroform, DMSO and water solutions. From the solvent-to-gas shift it is evident that higher solvent polarity increases the solute dipole moment and also stabilizes its structure. Moreover, taking as reference the energy of non-ionized (neutral) muscimol one can notice that the zwitterionic form is significantly less stable in the gas phase and polar solvents (by about 52 and 12 kcal/mol). However, the NH form of muscimol seems to be of lower stability in the gas phase (by about 6 kcal/mol) and is slightly more stabilized in polar solvents (by about -0.8 kcal/mol) than the non-ionized form.

It is known that the CCSD(T) calculations[92,93] with large basis sets could produce chemical accuracy of 1 kcal/mol in case of energy and energy related parameters. However, in our case the significantly lower level of theory applied cannot precisely answer this delicate question. In other words, from our calculations it is only evident that the zwitterionic form of muscimol is significantly less stable in the gas phase and solution than the two other structures. Thus, we could safely estimate the 1 : 1 ratio of NH and non-ionized muscimol forms in polar solutions.

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Table 3 B3LYP/6-311++G(3df,2pd) calculated dipole moments (D) and total energies (a. u.) of three forms of muscimol in the gas phase and solution. For comparison are included magnitudes of solvent-to-gas shifts of energy and dipole moment as well as energies (kcal/mol) relative to the neutral form

Parameter	Gas	$CHCl_3$	DMSO	Water	Solvent-to-gas shift (in D and kcal/mol)		
					$CHCl_3$	DMSO	Water
			Neutral				
Dipole moment	1.4636	1.8917	2.1158	2.1284	0.4281	0.6522	0.6648
Energy	-416.0850884	-416.0927186	-416.0957508	-416.0959065	-4.79	-6.69	-6.79
			Zwitterion				
Dipole moment	16.917	22.184	23.980	24.073	5.267	7.063	7.156
Energy	-416.0024541	-416.0563107	-416.077685	-416.0787653	-33.80	-47.21	-47.89
			NH form				
Dipole moment	3.242	**4.082**	4.417	4.434	0.840	1.174	1.192
Energy	-416.08259681	-416.0929542	-416.0970742	-416.0972857	-6.50	-9.08	-9.22
			Relative energy				
Neutral	0.00	0.00	0.00	0.00			
Zwitterion	51.85	22.85	11.34	10.76			
E + ZPV	51.78	23.85	12.61	12.03			
NH form	1.56	-0.15	-0.83	-0.87			
E + ZPV	6.37	-0.16	-0.81	-0.84			

3. 2. Comparison of experimental and theoretical NMR parameters of muscimol

Initially, calculations of TMS shieldings in the gas phase and solution were performed for comparison of predicted chemical shifts of three forms of muscimol in the gas phase and solution with experiment conducted in diluted solution of DMSO. For brevity, the corresponding carbon and proton nuclear shieldings of TMS are gathered in Table S2. As expected, TMS is a good NMR reference molecule (insensitive to environment effects) and only negligible effect of solvent could be noticed from Table S2 (from 0.2 to 0.6 for C and from 0.002 to 0.006 ppm for H). However, the shieldings are very sensitive to the level of

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theory. Thus, calculations with 6-31G* instead of aug-cc-pVTZ-J basis set produce TMS carbon and proton shieldings in the gas phase higher by about 6 and 0.4 ppm, respectively.

The measured ^{1}H and ^{13}C {-^{1}H} NMR spectra of muscimol in DMSO are shown in Figs. 1A - 1B. For brevity, the proton decoupled and coupled carbon spectra, as well as the enlarged fragments of ^{13}C {+^{1}H} NMR spectra are included in Supplementary Material (Fig. S1-S2).

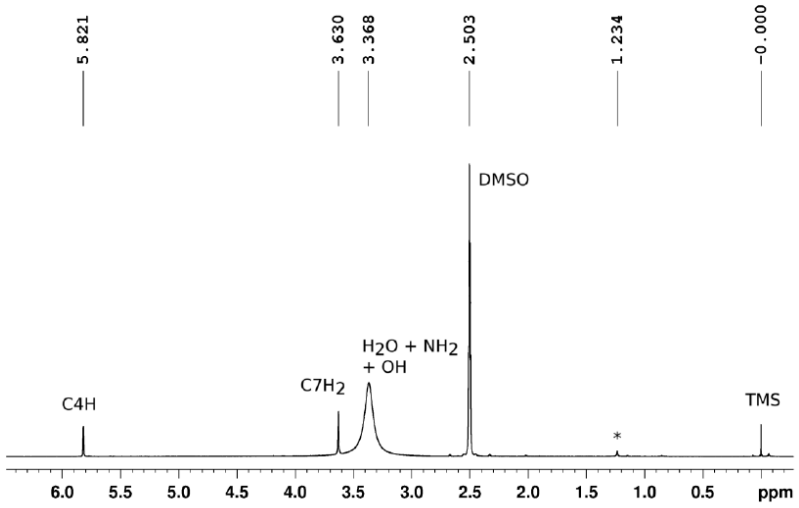

Fig. 1A. ^{1}H NMR spectrum of muscimol in DMSO-d6 (solvent impurity marked by star).

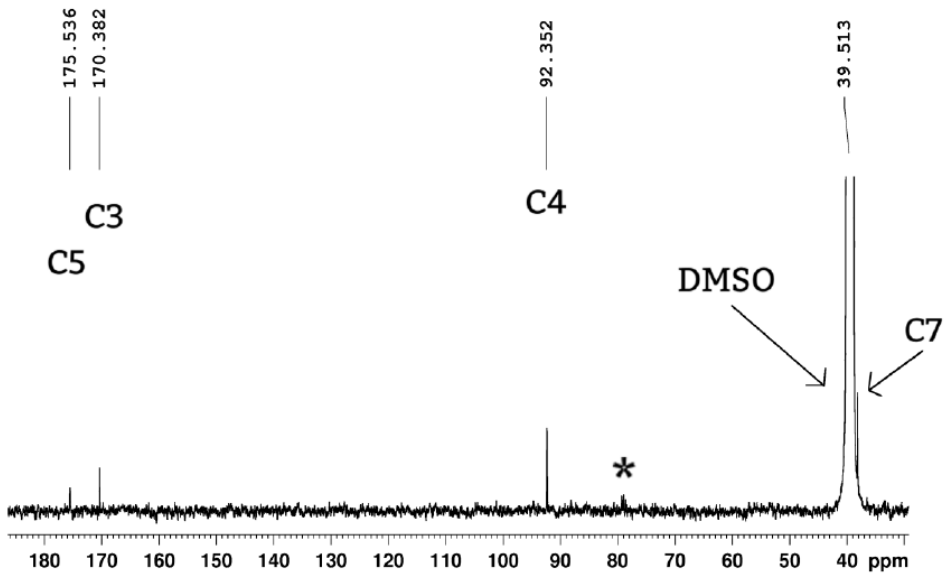

Fig. 1B. ^{13}C NMR spectrum of muscimol in DMSO-d6 (residual $CDCl_3$ impurity marked by star).

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There are several peaks visible in the proton spectrum (Fig. 1A). Starting from the most high-field TMS singlet (reference at 0.0 ppm), the next composite signal is a characteristic multiplet near 2.5 ppm (due to partly deuterated DMSO). At about 3.6 ppm is observed an intense and broad peak due to moisture in solvent, as well as overlapped signals from exchangeable protons (-OH, $-NH_2$ or $-NH_3^+$). It is important to note that this peak is at higher chemical shift than residual water in DMSO (3.3 ppm at room-temperature). The two single proton peaks from muscimol are recorded at about 3.6 and 5.8 ppm and their integral intensity ratio is 1 : 2. In the proton decoupled carbon spectrum (Fig. 1B), an intense multiplet at about 38.5 ppm, due to DMSO is visible, as well as four low intensity peaks originating from muscimol (at about 38, 92, 170 and 175 ppm). Their assignment is not possible from the $^{13}C\{-^{1}H\}$ NMR spectrum alone. However, looking at proton coupled carbon spectrum (Fig. S2) one can notice a splitting of the muscimol high-field signal into a doublet of nonequivalent intensity. In fact, we could expect the third, invisible component of multiplet being overlapped with solvent signal. Thus, the carbon signal at about 38 ppm is splitted into triplet and originates from a CH_2 group. The signal at about 92 ppm is from C4 since it splits into a doublet (J = 182 Hz) of triplets (J = 2.8 Hz). The signals at 170 and 178 ppm split into a doublet (J = 2.8 Hz) and the latter looks like an overlapped, irregular multiplet. The first signal should be due to C3 (splitted into doublet by the C4H proton) and the latter one due to C5 carbon. That signal indicates the presence of long-range couplings with several protons (here CH_2 and C4H).

The signal assignment in the ^{13}C NMR spectrum (Fig. 1B) was partly made by analyzing similar compounds and 2D proton-carbon correlation spectra (see Figs. S3-S4 in supplementary material for the corresponding HSQC and HMBC hetero-correlation spectra of muscimol in DMSO). From HSQC spectrum (Fig. S3) it is easy to notice a cross peak between a proton signal at about 3.6 ppm (singlet, 2H) and carbon signal at about 38 ppm, nearly overlapped with DMSO multiplet. Thus, this signal is due to CH_2 group. The low-field proton signal at about 5.8 ppm (singlet, 1H) is due to C4H ring fragment (it shows a cross-peak with a carbon peak near 92 ppm). A similar conclusion could be drawn from an HMBC spectrum in Fig. S4. Thus, the CH_2 protons are correlated strongly (via one bond) with carbon at about 37 ppm) and significantly weaker with signals at 92 and 175 ppm. In addition, the C4H proton is strongly correlated with a carbon signal near 92 ppm and weakly with two lowest-field carbon signals.

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In the next step we calculated nuclear shieldings of the three muscimol forms in the gas phase and solution using a fairly high level of theory. The equilibrium geometries in the gas phase and solution were calculated at B3LYP/6-311++G(3df,2pd) level of theory and the GIAO shieldings for all nuclei were predicted using an efficient basis set (aug-cc-pVTZ-J). This basis set was also used for calculation of indirect spin-spin coupling constants. The corresponding nuclear shielding values are gathered in Table 4. It is apparent that the calculated nuclear shieldings are sensitive to the molecular form of muscimol and solvent effect. However, analyzing the results from Table 4 it is obvious that the most sensitive techniques, able to distinguish between the three potential forms in solution, are ^{17}O and ^{15}N NMR spectroscopies[94-96]. In contrast to the carbon and proton data in these three forms, one could expect changes of muscimol oxygen and nitrogen shieldings of an order of 110 – 275 ppm! However, there is one important practical drawback in application of these techniques: larger samples (or isotopically labeled samples) are necessary to cope with intrinsic sensitivity problems for these nuclei. So, at this point we can only suggest running such experiments in the future.

The performance of B3LYP/aug-cc-pVTZ-J//B3LYP/6-311++G(3df,2pd) method in prediction of theoretical carbon and proton chemical shifts of three forms of muscimol in the gas phase and solution with respect to experimental data in DMSO is shown in Table 5. One could notice very large deviations between C3, C4 and C5 chemical shifts of NH form in DMSO with experiment (14, 10 and 18 ppm) and for C3 and C5 signals in non-ionized form (11 and 15 ppm). In case of zwitterion form of muscimol larger deviations for C3 and C4 chemical shifts are calculated (20 and 20 ppm). The RMS values for carbon chemical shifts in NH, neutral and zwitterion forms of muscimol in DMSO are 11, 10 and 15 ppm. The corresponding RMS values for protons are 0.17, 0.12 and 0.47 ppm. Thus, the carbon and proton data in Table 5 support the coexistence of non-ionized and NH forms of muscimol in DMSO solution. However, one of the reviewers suggested comparing our theoretical data with old chemical shifts measured in water (see footnote to Table 5). We derived the corresponding deviations and RMS values for theoretical data in the gas phase, chloroform, DMSO and water with an old experiment. The RMS_C values calculated for NH (14.336, 15.500, 16.032, 16.060 ppm), neutral (13.230, 13.933, 14.242, 14.258 ppm) and zwitterionic forms of muscimol (13.913, 10.390, 10.375, 10.397 ppm) are significantly larger than for our results in DMSO for the two former systems. The RMS_H values calculated for NH (0.581, 0.461, 0.418, 0.416 ppm), neutral (0.379, 0.308, 0.299, 0.298 ppm) and zwitterionic forms of muscimol (0.243, 0.101, 0.083, 0.083 ppm) are larger than for our results in DMSO. In case

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of proton chemical shifts the calculated RMS values for experimental data in water also disfavor NH and neutral forms of muscimol. On the contrary, in case of zwitterionic form of muscimol these data seem to somehow slightly prefer its structure in water. However, to verify this hypothesis one needs to perform several additional NMR experiments in water.

Table 4 Isotropic nuclear magnetic shieldings of muscimol atoms in its neutral, zwitterion and NH forms calculated using B3LYP/aug-cc-pVTZ-J basis set in the gas phase, chloroform, DMSO and water[a]

Atom	gas	$CHCl_3$	DMSO	Water
		NH form		
O1	76.482	68.615	64.865	64.662
O6	-35.797	3.132	18.680	19.477
N2	28.311	28.542	28.481	28.476
N8	221.986	223.322	223.843	223.870
C3	2.528	-0.144	-1.117	-1.166
C4	81.126	80.923	80.867	80.864
C5	-7.991	-9.688	-10.382	-10.419
C7	137.453	138.003	138.185	138.194
C4H	26.394	26.230	26.172	26.169
N2H	23.886	23.362	23.164	23.154
CH_2	28.253	28.149	28.110	28.108
NH_2	30.771	30.580	30.503	30.500
		Neutral		
O1	-32.775	-28.576	-26.861	-26.773
O6	240.548	239.988	239.592	239.569
N2	-113.748	-106.935	-104.177	-104.034
N8	219.080	220.611	220.989	221.003
C3	3.643	2.671	2.262	2.240
C4	88.296	87.095	86.537	86.507
C5	-5.504	-6.844	-7.336	-7.361
C7	138.234	138.677	138.845	138.853
OH(H6a)	26.723	26.285	26.122	26.114
C4H	26.115	25.921	25.836	25.831
CH_2	28.083	28.017	27.995	27.993
NH_2	30.762	30.584	30.514	30.511
		Zwitterion		
O1	40.861	24.836	17.119	16.804
O6	67.752	119.683	137.219	138.075
N2	-171.505	-131.826	-119.774	-119.192
N8	207.029	207.565	207.625	207.678
C3	-1.168	-5.497	-6.920	-6.987
C4	63.813	69.998	71.442	71.512
C5	30.186	19.210	14.352	14.113
C7	132.003	136.192	137.829	137.899
C4H	25.725	25.937	25.919	25.917
CH_2	27.298	27.441	27.462	27.462
NH_3	26.307	26.849	26.814	26.810

a) at B3LYP/6-311++G(3df,2pd) calculated equilibrium geometries in the gas phase and solution.

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Table 5 Comparison of selected theoretical carbon and proton chemical shifts (in ppm) of muscimol in its NH, neutral and zwitterion forms calculated using B3LYP/aug-cc-pVTZ-J basis set in the gas phase, chloroform, DMSO and water[a] with available experimental data in DMSO.

Atom	**Chemical shift[b]**					Deviation from experiment			
	gas	$CHCl_3$	DMSO	Water	**Exp.[cd]**	gas	$CHCl_3$	DMSO	Water
				NH form					
C3	180.610	183.658	184.817	184.876	**170.38**	10.230	13.278	14.437	14.496
C4	102.012	102.591	102.834	102.847	**92.35**	9.662	10.241	10.484	10.497
C5	191.129	193.202	194.083	194.130	**175.54**	15.589	17.662	18.543	18.590
C7	45.685	45.512	45.516	45.517	**38.16**	7.525	7.352	7.356	7.357
C4H	5.371	5.530	5.587	5.590	**5.823**	-0.452	-0.293	-0.236	-0.233
CH_2	3.512	3.612	3.650	3.652	**3.629**	-0.117	-0.017	0.021	0.022
N2H	7.879	8.399	8.595	8.605					
NH_2	0.994	1.180	1.256	1.260					
RMS_C						**9.914**	**10.851**	**11.268**	**11.291**
RMS_H						**0.330**	**0.207**	**0.168**	**0.166**
				Neutral					
C3	179.495	180.843	181.439	181.471	**170.38**	9.115	10.463	11.059	11.091
C4	94.842	96.419	97.164	97.204	**92.35**	2.492	4.069	4.814	4.854
C5	188.642	190.358	191.037	191.072	**175.54**	13.102	14.818	15.497	15.532
C7	44.904	44.837	44.856	44.858	**38.16**	6.744	6.677	6.696	6.698
C4H	5.650	5.840	5.923	5.928	**5.823**	-0.173	0.017	0.100	0.105
CH_2	3.682	3.744	3.764	3.766	**3.629**	0.053	0.115	0.135	0.137
OH(H6a)	5.042	5.476	5.637	5.645					
NH_2	1.003	1.177	1.245	1.248					
RMS_C						**8.752**	**9.877**	**10.373**	**10.400**
RMS_H						**0.128**	**0.082**	**0.119**	**0.122**
				Zwitterion					
C3	184.306	189.011	190.621	190.698	**170.38**	13.926	18.631	20.241	20.318
C4	119.325	113.516	112.259	112.199	**92.35**	26.975	21.166	19.909	19.849
C5	152.952	164.304	169.349	169.598	**175.54**	-22.588	-11.236	-6.191	-5.942
C7	51.135	47.322	45.872	45.812	**38.16**	12.975	9.162	7.712	7.652
C4H	6.040	5.824	5.840	5.842	**5.823**	-1.356	-1.504	-1.526	-1.526
CH_2	4.467	4.320	4.297	4.297	**3.629**	1.829	1.283	1.316	1.320
NH_3	5.458	4.912	4.945	4.949					
RMS_C						**20.001**	**15.853**	**15.032**	**15.005**
RMS_H						**0.593**	**0.488**	**0.472**	**0.472**

a) Using B3LYP/6-311++G(3df,2pd) equilibrium geometries in the gas phase and solution;
b) Chemical shifts were calculated using TMS data from Table S2.; c) this work in DMSO; d) old experimental data[10,11] in D_2O significantly differ from our values (178.1, 100.1 164.4, 35.8 ppm for carbons and 5.85 and 4.18 ppm for protons).

In order to compare our theoretical shieldings of three forms of muscimol with experiment performed at room temperature one needs to include rovibrational and temperature

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corrections. Because of very demanding calculations we started with VPT2 method and a fairly small basis set (6-31G*) in the gas phase and DMSO solution. For brevity, these theoretical nuclear shieldings are compared with experimental data before and after inclusion of ZPV and TC corrections in the supplementary material (see Tabs. S3 – S5). Besides, these tables include deviations of theoretical nuclear shieldings from experimental and empirical values and the corresponding carbon and proton RMS values. In Table 6 we compared the RMS values to get an overview of the agreement between theoretically predicted proton and carbon nuclear shieldings using B3LYP/6-31G* calculations, additionally augmented with zero-point vibration and temperature corrections, and experimental data in DMSO. It is apparent from Table 6 that carbon nuclear shieldings in vacuum are calculated with lower accuracy than in DMSO (RMS of about 7 and 5 ppm for NH form before rovibrational correction). Besides, the inclusion of ZPVC+TC improves the agreement with empirical carbon values (RMS drops for the NH form from about 5.1 to 3.7 ppm). The ^{13}C RMS values (shown in parenthesis for empirical data) are getting better in the following order: NH form > neutral > zwitterion (14 > 5.6 >3.7). On the other hand, it is important to note a slight worsening of proton data upon inclusion of solvent and rovibrational data. For example, for NH form of muscimol the RMS of proton signals in vacuum and DMSO is 0.75 and 0.86 ppm, respectively. The quality of proton signal prediction (RMS of 0.9, 1.0 and 1.2ppm) deteriorates in the same order (NH form > neutral > zwitterion). Thus, the data in Table 6 indicate a somehow better agreement with experiment for NH model than for the neutral muscimol and exclude the presence of its zwitterionic form in DMSO solution. This conclusion is supported by our energy calculations gathered in Table 3 (keep in mind a small predominance of NH form over neutral muscimol and a significantly higher energy of the zwitterion). In Table S5 we also compared old results in water[10,11] with our theoretical predictions additionally augmented with rovibrational corrections. These data clearly showed about 2-3 times smaller RMS values for our theoretical results compared with our measured data in DMSO (see also Table S4). Hence, probably we should treat with caution these results[10,11].

The performance of GIAO NMR calculations on muscimol nuclear shieldings seems to be of low quality. However, similar size of deviations (and RMS values) between theoretical and experimental carbon and proton nuclear shieldings were recently reported[97] in case of two studied dehydroamides in DMSO. The systematic improvement and control of our DFT results could be sometimes questionable[71]. At this point we could only remind the semi-empirical nature of density functionals, optimized for energy, and fortuitous error

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cancellations for some combination of method and basis set. The simplified introduction of solvent effect and rovibrational corrections also could not accurately describe the system. However, the use of higher level theory, for example coupled clusters for the systems of this size is still very expensive[71].

On the other hand, as suggested by one of the reviewers, one could also use a more pragmatic approach to get a reasonably (or better) agreement between theory and experiment. Therefore, using the "optimal" wave function and basis set size and quality for computational efficiency, as well as supported by additional regression or statistical manipulations seems to be adequate in many cases[98-100]

Indeed, sometimes by selecting various statistical data processing methodologies or simple Hartree-Fock calculations without electron correlation and the smallest basis set (3-21G) one could nicely reproduce experiment. In many cases such a result is due to favorable accidental error cancellation. However, this approach is not general and it works for well behaving and "similar" molecular systems.

Table 6 Comparison of RMS values (in ppm) of carbon and proton nuclear shielding of three forms of muscimol in the gas phase and DMSO calculated using B3LYP/6-31G* method from experimental results in DMSO[a]

RMS	**Exp.**	**Emp.[b]**	**Exp.**	**Emp.[b]**
	Vacuum		DMSO	
		NH form		
C	7.03	5.15	5.23	3.74
H	0.36	0.75	0.28	0.86
		Neutral		
C	9.44	6.64	8.40	5.62
H	0.39	0.94	0.45	1.02
		Zwitterion		
C	22.82	21.02	15.43	14.22
H	0.83	1.40	0.77	1.18

a) TMS experimental shieldings are 186.37 and 32.815 ppm for C and H, from ref. [86,87]; b) Rovibrational corrections (calculated at the same level of theory) are included.

In the final step we analyzed the last type of available data from NMR experiment in DMSO: the observed carbon-proton indirect spin-spin coupling constants through n bonds ($^{n}J_{CH}$ or SSCC). In Table 7 are gathered experimental carbon-proton couplings and selected theoretical parameters (B3LYP/aug-cc-pVTZ-J//B3LYP/6-311++G(3df,2pd) calculated in the gas phase and solution). In addition, in the supplementary material are shown spectral

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fragments showing the corresponding $^{n}J_{CH}$ patterns (Fig. S1-S2). Obviously, two very large one-bond $^{1}J_{CH}$ couplings of about 182 and 148 Hz are visible in the spectra (doublet and triplet patterns assigned to carbons C4H and CH_2). Besides, significantly smaller signal splitting (2 – 3 Hz) of some signals is clearly visible. For completeness, we also showed some couplings via several bonds which could be potentially visible in the spectra. However, it was difficult to observe these splitting in the recorded spectra. For example, the C5 signal appeared as very complex pattern, formed by irregularly overlapped peaks. So, it was very difficult to analyze significantly overlapped small signals of this composite peak.

Analyzing the theoretical data in Table 7 we notice a fairly small effect of solvent. For example, the gas-to-DMSO shift for ^{1}J(C4H) calculated for NH, non-ionized and zwitterion forms of muscimol are 2.3, 3.5 and 0.8 Hz, respectively. This is only about 1.3, 1.9 and 0.4% of the observed coupling of 182.4 Hz in DMSO. Similarly, this coupling calculated in the gas phase for the three forms of muscimol (194.7, 192.7 and 187.8 Hz) differs by about 6.1 Hz (or 3.3% of the experimental values) and slightly more in DMSO (197.0, 196.2 and 188.6 Hz makes the largest difference of 8.4 Hz or 4.6% of the measured parameter). The deviation of theoretically calculated ^{1}J(C4H) coupling in DMSO for the NH, neutral and zwitterion forms of muscimol from experiment is 14.6, 13.8 and 6.2 Hz. Thus, this level of theory overestimates the experimental value by about 8.0, 7.6 and 3.4%. In case of one bond carbon-proton coupling in CH_2 group, the theoretical value differs by about -0.9, -2.0 and 9.4 Hz, respectively. These deviations are smaller (0.6, 1.4 and 6.4% of the experimental coupling). Unfortunately, the presence of small coupling via several bonds makes the comparison of theory with experiment less reliable and the calculated RMS values for deviations in DMSO (6.68, 6.22 and 5.03 Hz) seem to be within 20% (6±1 Hz). Thus, the carbon-proton couplings cannot be used to distinguish between the three studied forms of muscimol in DMSO. Can one improve the accuracy of theoretical prediction of SSCC parameters[101]? The rovibrational corrections could in principle improve the agreement between theory and experiment. However, in comparison to nuclear shieldings, it is much more difficult to calculate the correction to the coupling constants[101]. Besides, in our opinion, these corrections could consistently improve results obtained with the more advanced theoretical methods (SOPPA[102] or EOM CCSD[103,104]) but not those, obtained from DFT approach[101], which is somehow semi-empirical. On the other hand, one should keep in mind that the errors in SSCC values for muscimol fall within the 10-20% range of deviations from empirical values for a set of several small molecules, calculated with selected density functionals in the complete basis set limit

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(the results obtained with aug-cc-pVTZ-J basis set were also fairly similar) in the gas phase[101].

At the very end of our study we would like to discuss briefly the changes in atomic charges on the three forms of muscimol in the gas phase and solution. In this case the most polar solvent - water was used. Obviously, Mulliken charges directly produced in standard Gaussian calculations are believed to be the least reliable and very sensitive to method of calculation and basis set completeness. This was nicely demonstrated in case of water charges by Martin and Zipse[105]. They noticed significantly better performance of Merz-Kollman scheme[106,107] in calculation of electrostatic potential derived charges (ESP). In Figs. S5 and S6 we plotted the Mulliken and ESP charges of "heavy atoms with hydrogen summed" for three forms of muscimol in the gas phase and water. For completeness, these summed charges and charges for all atoms are placed in Tabs. S6-S7 in the supplement. It was pleasing to notice similar qualitative changes in charges going from neutral to NH and zwitterion forms of muscimol (Figs. S5-S6). Obviously, the magnitude of charges obtained with both methods differs. These figures clearly show the decrease of charge at O2 for NH and zwitterion forms of muscimol and increase of charge on N2 for NH form, as well as an increase at N8 for the zwitterion (see also arrows indicating the biggest changes). All these changes of charges are in agreement with our expectations (for example, N8 is positively charged upon protonation and O6 in zwitterion is negative charged). Our results on muscimol charges are in agreement with earlier results obtained with smaller basis set by Serdaroglu[58].

Conclusions

Structural and NMR studies on muscimol, an agonist of GABA from fly agaric mushroom are reported. Up to now, these are the most theoretically and experimentally advanced studies which could improve the understanding of basic features of muscimol at atomistic level. Theoretical DFT and experimental NMR studies were conducted to explain the coexistence of NH and non-ionized (neutral) forms of muscimol in DMSO solution. The B3LYP/6-311++G(3df,2pd) calculated structural parameters of zwitterionic form of muscimol in polar solvents using PCM model resemble the earlier reported crystal structure. The two other, potential forms of muscimol were also considered in our calculations: non-ionized (neutral) and NH. The total energies of these forms were roughly identical in polar solvents and the zwitterionic form was significantly less stable (by about 12 kcal/mol).

GIAO NMR calculations of nuclear shieldings (and chemical shifts) at B3LYP/6-31G* and B3LYP/aug-cc-pVTZ-J levels of theory supported assignment of experimental ^{1}H and ^{13}C

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NMR spectra of muscimol in DMSO. The rovibrational effects to calculated nuclear shieldings were also included using VPT2 methodology.

From the structural and energetic parameters, obtained from unconstrained geometry optimization in the gas phase and solution, the existence of equal amounts of non-ionized and NH forms in polar solvents is deduced. The zwitterionic form is significantly less stable in solution and the gas phase (by 12 and 52 kcal/mol). This conclusion was supported by comparing theoretical carbon and proton nuclear magnetic shieldings with measured NMR spectra in DMSO. These results could also help in understanding biological action of GABA analogs. Besides, we want to suggest future experimental ^{17}O and ^{15}N NMR studies as the most sensitive techniques for distinguishing different forms of muscimol in solution.

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Table 7 B3LYP/aug-cc-pVTZ-J//B3LYP/6-311++G(3df,2pd) calculated indirect spin-spin coupling constants (in Hz) for three forms of muscimol in the gas phase and solution. Experimental values observed from $^{13}C\{+^{1}H\}$ NMR spectra of muscimol in DMSO and the deviations of theoretical results from experiment are given, too.

SSCC	Theoretical				**Exp.**	Deviation from Exp.			
	Gas	$CHCl_3$	DMSO	Water	**in DMSO**	Gas	$CHCl_3$	DMSO	Water
				Muscimol NH					
^{1}J(C4H4a)	194.684	196.406	197.022	197.052	**182.411**[a]	12.273	13.995	14.611	14.641
^{1}J(C7H7aver)	145.102	146.498	147.035	147.062	**147.970**	-2.868	-1.472	-0.935	-0.908
^{2}J(C3H4a)	5.615	5.890	6.035	6.043	**3.143**	2.472	2.747	2.892	2.900
^{2}J(C5H4a)	12.452	11.915	11.621	11.605	**2.317**[b]				
^{2}J(C5H7aver)	-1.782	-1.955	-2.072	-2.079					
^{3}J(C4H7aver)	2.985	3.006	3.005	3.005	**2.779**	0.206	0.227	0.226	0.226
^{3}J(C7H4a)	1.454	1.380	1.348	1.347					
^{4}J(H4aH7aver)	-0.417	-0.391	-0.380	-0.380					
RMS						**5.745**	**6.413**	**6.675**	**6.688**
				Muscimol neutral					
^{1}J(C4H4a)	192.720	195.110	196.184	196.242	**182.411**[a]	10.309	12.699	13.773	13.831
^{1}J(C7H7aver)	144.346	145.553	146.009	146.031	**147.970**	-3.624	-2.417	-1.962	-1.939
^{2}J(C3H4a)	2.693	2.885	2.978	2.983	**3.143**	-0.451	-0.258	-0.165	-0.160
^{2}J(C5H4a)	11.075	10.911	10.805	10.799	**2.317**[b]				
^{2}J(C5H7aver)	-2.151	-2.324	-2.445	-2.452					
^{3}J(C4H7aver)	2.898	2.940	2.959	2.960	**2.779**	0.119	0.161	0.180	0.181
^{3}J(C7H4a)	0.863	0.893	0.901	0.901					
^{4}J(H4aH7aver)	-0.433	-0.413	-0.395	-0.394					
RMS						**4.891**	**5.783**	**6.223**	**6.247**
				Muscimol Zwitterion					
^{1}J(C4H4a)	187.846	187.644	188.583	188.649	**182.411**[a]	5.435	5.233	6.172	6.238
^{1}J(C7H7aver)	158.234	157.621	157.346	157.334	**147.970**	10.264	9.651	9.376	9.364
^{2}J(C3H4a)	2.702	3.485	3.938	3.962	**3.143**	-0.441	0.342	0.795	0.819
^{2}J(C5H4a)	14.140	12.299	11.605	11.568	**2.317**[b]				
^{2}J(C5H7aver)	-3.915	-4.135	-4.181	-4.184					
^{3}J(C4H7aver)	2.007	2.297	2.524	2.536	**2.779**	-0.772	-0.482	-0.255	-0.243
^{3}J(C7H4a)	1.066	1.157	1.164	1.166					
^{4}J(H4aH7aver)	-0.937	-0.711	-0.577	-0.573					
RMS						**5.209**	**4.917**	**5.034**	**5.046**

a) determined as 182.200 Hz from ^{1}H NMR spectrum in DMSO (as distance between carbon-proton satellites) and 182.411 Hz from $^{13}C\{+^{1}H\}$ NMR spectrum in DMSO; b) It differs significantly from the theoretical values. However, this value is only roughly guessed from the number of partly overlapped peaks, additionally buried with noise.

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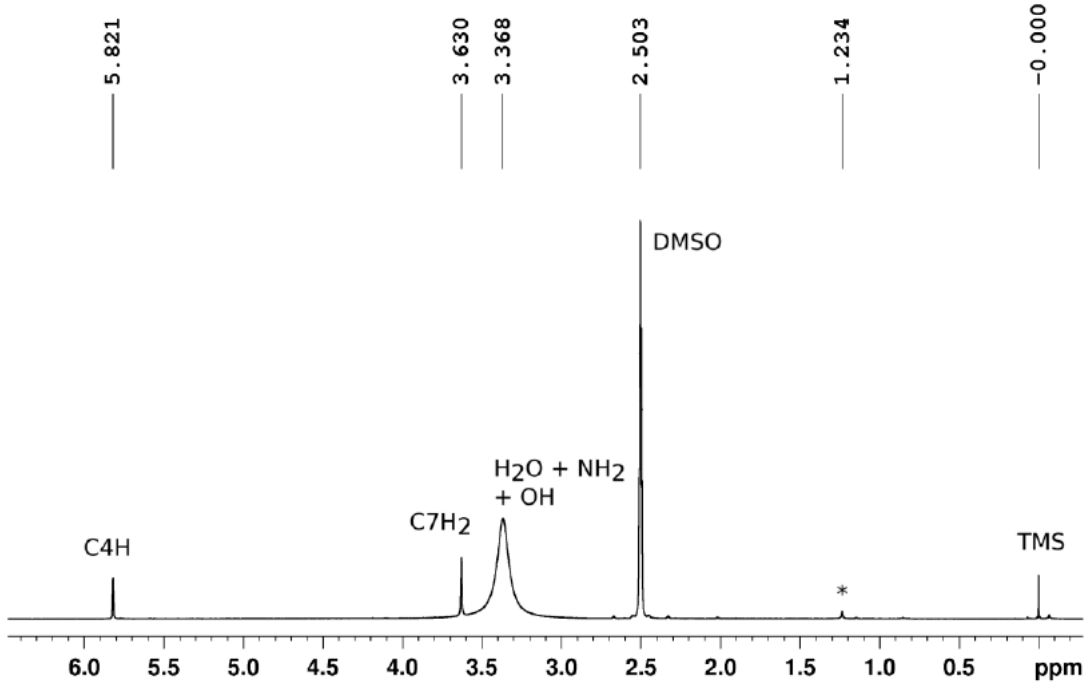

Fig. 1a

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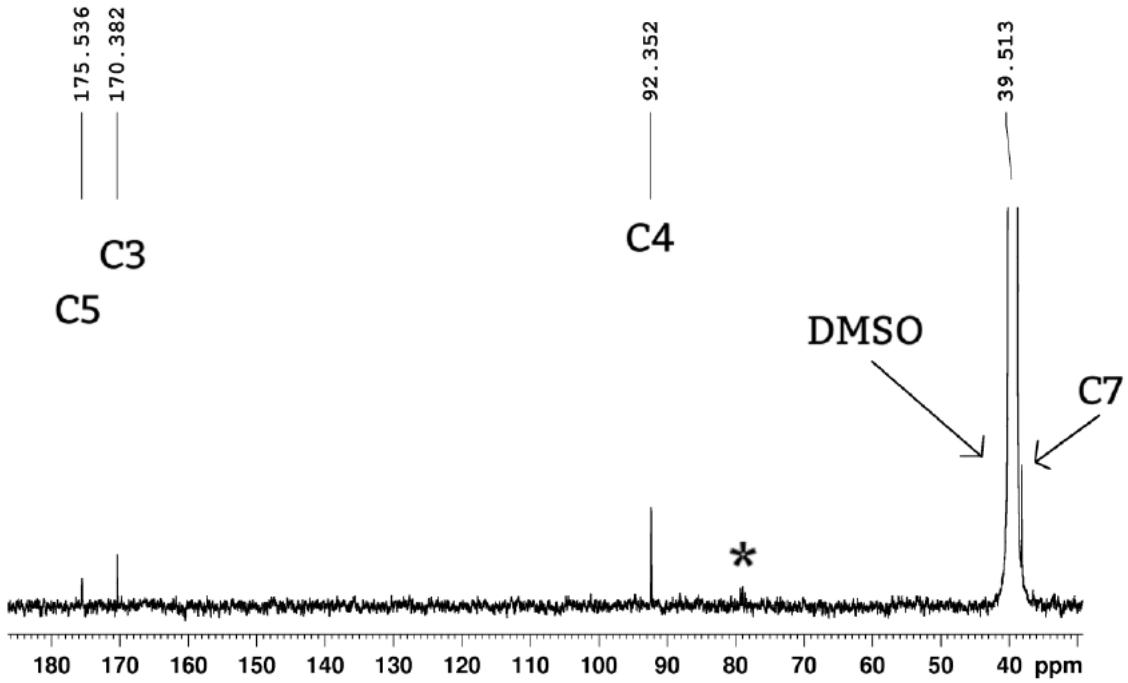

Fig. 1b

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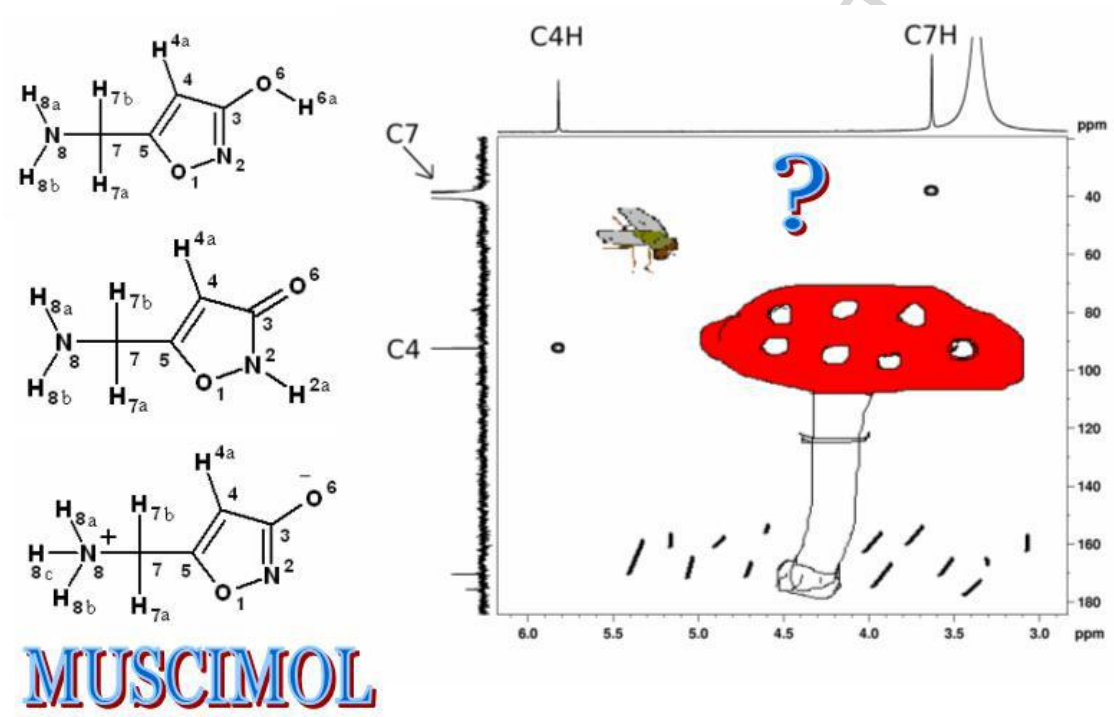
Theoretical and experimental NMR studies on muscimol from fly agaric mushroom (*Amanita muscaria*)

Teobald Kupka and Piotr P. Wieczorek



Graphical abstract

High level theoretical studies support the NMR experiment and provide new evidence on the form of GABA agonist - muscimol in DMSO solution.

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Highlights

- NMR and ab initio/DFT studies on muscimol are proposed.
- The use of high level theoretical modeling could aid interpretation of muscimol structure.
- The observed NMR data of muscimol in DMSO are assigned.
- The non ionized and NH- forms of muscimol (1 : 1) in DMSO are proposed.

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